

CUSTOMER RETENTION STRATEGY • BUSINESS ANALYTICS CASE STUDY

Who won't come back, and where retention spend actually pays off

A churn-prediction and value-based targeting analysis of
232,079 shoppers across five months of a cosmetics store

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PART I

Business Context

- Executive summary
- Business problem
- Project approach



Most first-time shoppers never return, and 98% of the revenue at risk sits in just two segments

77%

of October shoppers do not
return within 60 days

0.82

ROC-AUC churn model, beating
a recency-only rule (0.74)

\$572k

total 60-day revenue at risk -
70% in one segment

\$0.05

spend per \$1 defended under
value-first targeting, vs \$99
churn-first

Recommendation: shift retention budget from the highest-churn segment to the highest revenue-at-risk segment, and act on early week-one behaviour before customers are gone.

On a browse-heavy store, value comes from repeat visits - yet three in four shoppers vanish

Situation

A cosmetics store where low-ticket, impulse purchases mean revenue depends on customers coming back, not on any single order.

Complication

77% of October-active users never return in the next 60 days. Retention, not acquisition, is the binding constraint.

Question

Who will not return? Which early behaviors predict that? And where should retention budget be spent?

A leakage-free model built and validated to withstand scrutiny

Data

232k users • 5 months of view / cart / purchase events;
October behaviour, Nov-Dec outcomes

Leakage-free label

Features from October only; churn measured from a disjoint
future window - recency can't leak the target

Two models

Interpretable logistic regression benchmarked against gradient
boosting for the predictive ceiling

Validation

5-fold cross-validation (± 0.001), naive baselines, permutation
importance, calibration check

PART II

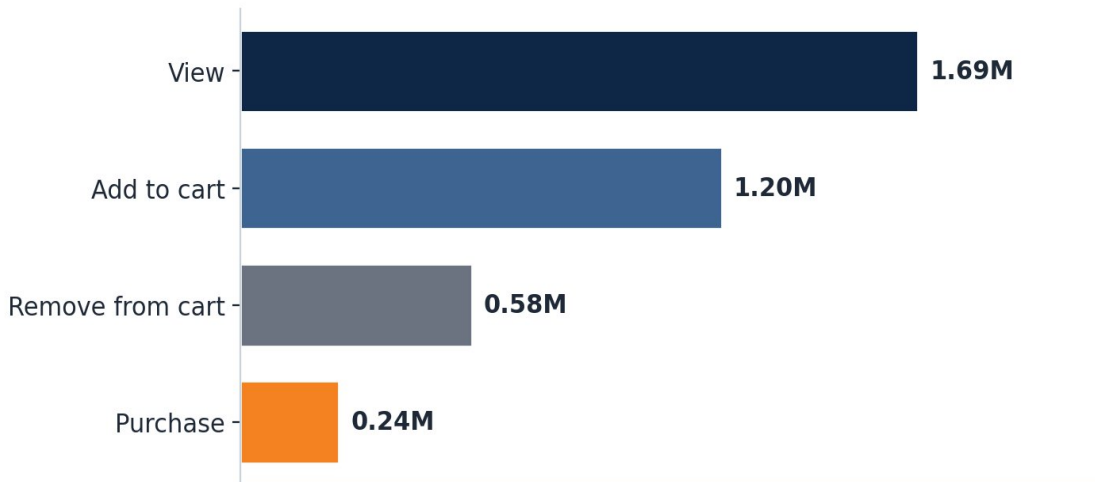
Business Diagnostics

- Understanding customer behaviour
- Funnel analysis
- Retention analysis



A high-volume, low-ticket, browse-heavy store

Event mix across 3.7M cleaned October interactions



3.7M

Cleaned interactions from 232,079 shoppers

\$4.11

Median item price - impulse-priced

25,755

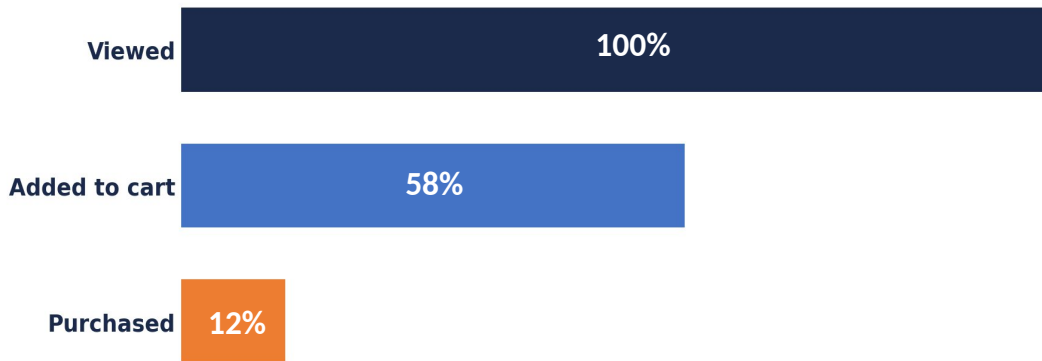
Buyers; median buyer spend \$32

1 session

For the median user - mostly one-and-done

58% of viewers add to cart - but only 1 in 5 carts convert

Conversion funnel: the cart→purchase step is the biggest leak



Where the leak is

The steep drop from intent to purchased is the biggest single leak in the funnel: a large share of shoppers add products to the cart but never complete a purchase.

A cart-stage recovery mechanism has the largest surface area to lift returns and revenue.

Most users never form a habit - retention collapses right after the first visit

1

session for the median user, most shoppers appear once and leave

Week 0

is where the cohort curve falls off a cliff, few return the next week

Early

depth (multiple sessions, more active days) is the strongest protective signal

Churn must be predicted from week-1 behaviour. By the time inactivity is visible, the customer is already gone, which is also why a naive 30-day-inactivity label leaks.

PART III

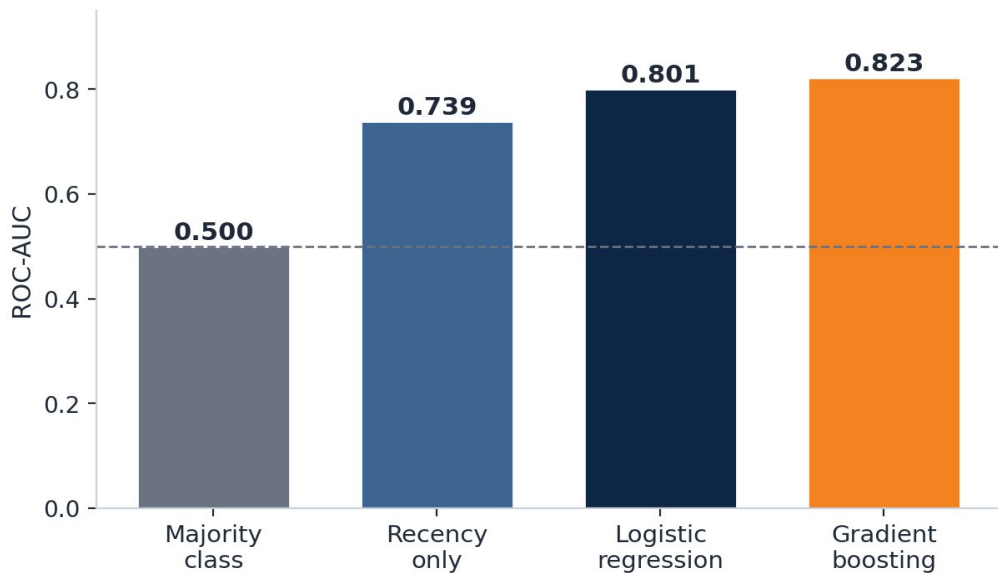
Predictive Analytics

- Predicting customer churn
- What drives customer churn?



The behavioral model predicts non-return at ROC-AUC 0.82, clearly beating recency alone

Predictive lift over naive baselines (5-fold CV)

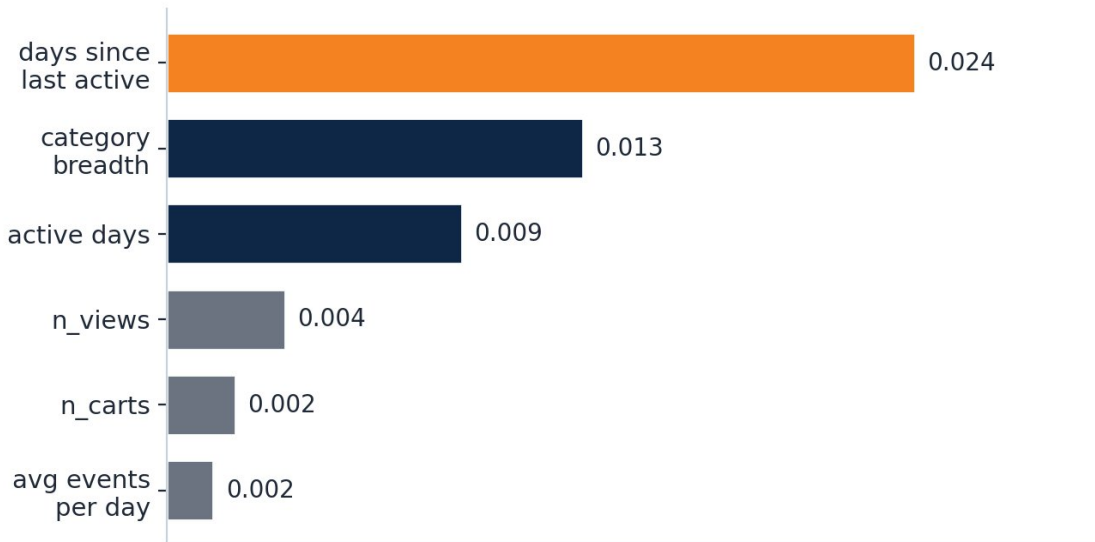


What it means

Recency alone delivers strong predictive performance (ROC-AUC = 0.74). Adding customer behavior features further improves performance to 0.80-0.82. Because non-return accounts for 77% of customers, ROC-AUC and the improvement over the 0.50 baseline provide a more reliable measure of model performance than PR-AUC.

Three early signals separate returners from churners

What predicts non-return (permutation importance)



What it means

Recency within October is the top risk signal, category breadth and active days are associated with lower non-return risk, and both models agree, above their error bars. Shoppers who explore broadly and return across days are far likelier to stay.

PART IV

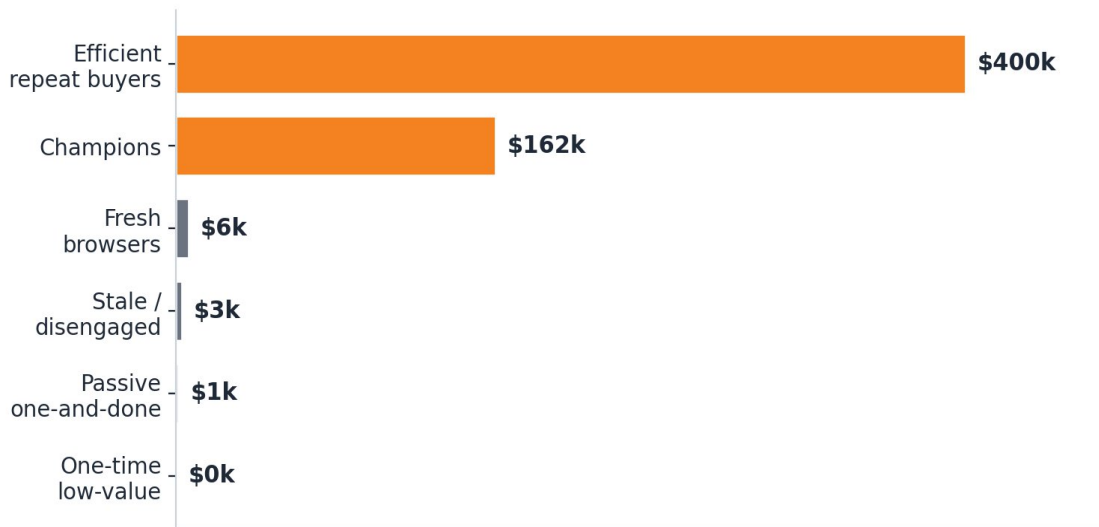
Business Strategy

- Where is revenue actually at risk?
- Value-based targeting strategy
- Financial impact



Churn rate misleads: 98% of the \$572k at risk sits in just two segments

Revenue at risk by customer segment (60-day horizon)



\$400k

sits with **Efficient repeat buyers** - more than the other five segments combined. Champions add \$162k. The 91%-churn stale segment carries just \$3k.

Contact a user only when churn probability × incremental uplift × value exceeds contact cost

The model produces well-calibrated probabilities (Brier Score = 0.129), making its predictions reliable for business decisions. Customers are contacted only when their expected value exceeds the intervention cost (break-even: \$22.50). This ensures marketing resources are focused on high-value, at-risk customers while avoiding unnecessary outreach to low-value segments.

3.4%

Of users selected - a focused, fundable campaign

\$81 vs \$2.56

Average Oct spend of targeted vs non-targeted

37% / 22%

Of repeat buyers and champions targeted, 0% of stale

Value-first targeting defends \$1 of revenue for 5 cents; churn-first pays \$99

Churn-first (naive)

Contact the whole 91%-churn stale segment

Users contacted	111,339
Campaign spend / cycle	\$301k
Revenue at risk addressed	\$3k
Campaign cost per \$1 of at-risk revenue addressed	\$98.8

Value-first (recommended)

Contact only users with positive expected value

Users contacted	~7,960
Campaign spend / cycle	\$21k
Revenue at risk addressed	\$400k segment
Campaign cost per \$1 of at-risk revenue addressed	\$0.05

Expected profit ≈ \$21k per 60-day cycle at full user base under illustrative economics (\$2.70/contact, 12% uplift). A fixed 15% budget rule tests negative (−\$29k) at the same scale: only 3.4% of users clear the break-even, so the quota forces ~8,000 additional contacts below it, each destroying about \$1.87, so the EV>0 rule is retained. Dollar inputs are assumptions to tune against real campaign costs.

PART V

Implementation

- Strategic recommendations
- Implementation roadmap
- Validation plan
- Limitations & next steps



Three levers, sequenced by return on effort

01

48-hour cart-abandonment nudge

Recover the largest funnel leak while intent is warm.

Metric: cart→purchase, 60-day return

02

Early-activation onboarding

Build week-1 depth for shallow, single-session users.

Metric: sessions/user, week-1→2 retention

03

High-value win-back program

Protect the ~\$400k at-risk repeat-buyer segment first.

Metric: churn in high-value tier

From model to money in three phases

PHASE 1

Prove

Weeks 1–4

- Deploy churn risk score to CRM
- Launch cart-nudge A/B on the \$400k segment
- Baseline dashboards: return rate, cart→purchase

KPI: A/B live • risk score in CRM

PHASE 2

Scale

Months 2–3

- Read out A/B; roll out winning nudge
- Launch early-activation onboarding flow
- Wire EV>0 value-first targeting into campaigns

KPI: +3pp 60-day return; +5pp cart→purchase

PHASE 3

Institutionalise

Months 4–6

- Monthly model retrain + drift monitoring
- Quasi-experimental causal read (propensity-score matching / difference-in-differences)
- Extend label windows; upgrade spend to CLV

KPI: stable AUC ≥ 0.80 ; +\$20k / cycle profit

Gate to scale: A/B shows ≥ 3 pp lift in 60-day return at 95% confidence, with flat opt-out and revenue guardrails.

Prove it before scaling: a powered, guardrailed A/B test

Note: the dataset is observational - this is a validated experimental design, not a completed A/B test.

Hypothesis

A 48-hour cart-abandonment nudge lifts the 60-day return rate

Design

Users randomised 50/50; two-proportion z-test on return rate with a 95% CI

Power

Sized for a 3pp lift at 80% power from the 22.9% baseline return rate

Guardrails

Notification opt-out and revenue/user; fixed sample size - no peeking

What this analysis is - and is not

Limitations

- “Churn” means non-return, not subscription cancellation
- No demographics or marketing channel - product/UX levers only
- Nov-Dec label overlaps Black Friday, so churn is a lower bound
- Dollar targeting inputs are illustrative assumptions

Next steps

- Run the cart-nudge A/B test on the priority segment
- Deploy calibrated risk scores for live targeting
- Extend the cohort view across all five months
- Validate causal lift with matching / difference-in-differences

FINAL TAKEAWAY

Predict non-return early, then spend where revenue is at risk, not where churn is highest.

Rose Nguyen · Customer Retention Strategy

